Huawei Access Node Optical Transport Failure Resolution (ALARM_3740)

Node: WHI-ACC-81 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_3740

Description

The Optical Transport Failure in the Huawei Access Node WHI-ACC-81 at Whitefield East POP is attributed to the degradation of SFP/XFP transceivers, fiber cuts, or Rx/Tx power loss. This failure domain occurs due to the inherent vulnerabilities of the optical transport protocol in Huawei's Access Node equipment. The Optical Transport Network (OTN) operates on the control plane, utilizing the Generalized Multi-Protocol Label Switching (GMPLS) protocol to manage the optical path. However, the reliability of OTN is heavily dependent on the physical layer, where the SFP/XFP transceivers and fiber optic cables are susceptible to degradation and cuts. Furthermore, the Rx/Tx power loss can be caused by the attenuation of the optical signal due to the length and quality of the fiber optic cables. In the event of an Optical Transport Failure, the Huawei Access Node WHI-ACC-81 will raise a critical alarm (ALARM_3740) to notify the network administrators of the failure.

Pre-requisite Checks

- Pre-Check 1: Verify the site location and node ID to confirm the affected device (WHI-ACC-81 at Whitefield East POP).

- Pre-Check 2: Check the redundant link status to ensure that the failure is not due to a redundant link failure.

- Pre-Check 3: Dispatch remote hands to the Whitefield East POP to ensure access to the affected device.

Potential Root Causes

Cause	Probability
SFP/XFP transceiver degradation	High
Fiber cut	Medium
Rx/Tx power loss due to fiber optic cable degradation	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical transceiver, Expected Output: Transceiver Type: SFP+/XFP, Temperature: 25°C, Voltage: 3.3V, Description: This command displays the type and status of the SFP/XFP transceiver, including temperature and voltage readings. It helps to verify if the transceiver is functioning correctly and if the failure is due to hardware or software issues.

Command: display optical interface, Expected Output: Interface: OTHR-1-1-1, Status: Active, Tx Power: -20dBm, Rx Power: -30dBm,
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```
Description: This command displays the status and power levels of the optical interface, which helps to identify if the failure is due to Rx/Tx power loss or fiber cut.

Command: display fiber status, Expected Output: Fiber Status: Cut, Fiber ID: FIB-1-1-1, Description: This command displays the status of the fiber optic cable, which helps to verify if the failure is due to a fiber cut.
```

Resolution Steps

Step 1: Step: Swap the SFP/XFP transceiver, Command: swap optical transceiver 1/1/1, Description: Swap the SFP/XFP transceiver with a new one to rule out hardware issues. Verify the transceiver type and status after swapping.

Step 2: Step: Check and clean the fiber optic cable, Command: display fiber status, Description: Verify the fiber status and clean the fiber optic cable if it's dirty or damaged. Check the fiber status again after cleaning.

Step 3: Step: Adjust the Rx/Tx power levels, Command: set optical interface othr-1-1-1 tx-power -20dBm, Description: Adjust the Tx power level to -20dBm to ensure that the optical signal is within the acceptable range. Verify the Rx power level after adjusting the Tx power level.

Post-Check Verification

```
Command: display optical transceiver, Expected Output: Transceiver Type: SFP+/XFP, Temperature: 25°C, Voltage: 3.3V, Description: Verify that the SFP/XFP transceiver is functioning correctly and that the temperature and voltage readings are within the acceptable range.

Command: display optical interface, Expected Output: Interface: OTHR-1-1-1, Status: Active, Tx Power: -20dBm, Rx Power: -30dBm, Description: Verify that the optical interface is active and that the Tx and Rx power levels are within the acceptable range.

Command: display fiber status, Expected Output: Fiber Status: Normal, Fiber ID: FIB-1-1-1, Description: Verify that the fiber optic cable is functioning correctly and that the fiber status is normal.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: swap optical transceiver 1/1/1, Description: Restore the original SFP/XFP transceiver to its original position to rule out any hardware issues.

- Step: Restore the original Rx/Tx power levels, Command: set optical interface othr-1-1-1 tx-power original, Description: Restore the original Rx/Tx power levels to their original values to ensure that the optical signal is within the acceptable range.